

Purification of nucleic acids from agarose gels

Recovery of DNA fragments from agarose gels is necessary when undesired side-products cannot be removed or inactivated ahead of sensitive downstream applications such as cloning or sequencing.

The DNA can be electro-eluted into a trough excised ahead of the band (McDonnell et al., 1977), or more conveniently, the band can be excised directly.

To remove the agarose, fast flush extraction (Grey and Brendel, 1992) (*Alternative A*) or chaotropic salts such as guanidinium thiocyanate (Matitashvili and Zavizion, 1997) (*Alternative B*) are commonly used. The latter, as a silica-based method, being convenient, at the expense of lower recovery and higher cost. Fast flush extraction is thus preferred for large sample sets.

Risk assessment

- Transilluminators emit ultraviolet radiation, which can damage unprotected eyes and skin.
- Guanidine salts form TOXIC GASES with bleach or strong acid
- ▷ Wear gloves, safety glasses, lab coat
- DO NOT add bleach or acidic solutions directly into sample waste



Reviewed: Nov 26, 2023

Procedures

» Separation and excision of the desired DNA fragment from agarose gels

- (1.) Resolve the desired DNA fragment on a 0.6–1.2% agarose gel in TAE or lithium acetate buffer. Load at least 200 ng per lane. 

Critical: To avoid carry-over of DNA, use fresh running buffer and leave empty lanes between samples. 

Critical: Avoid TBE buffer if possible, since borate inhibits downstream enzymatic reactions. 

- (2.) Visualize the gel using a blue light or long-wave UV source in a darkroom or gel box.

Critical: Always use long-wave UV (> 400 nm) and minimize exposure. Avoid pre-imaging gels under UV to prevent DNA damage and pyrimidine dimer formation. 

- (3.) Excise the desired band with a clean surgical blade. Trim excess agarose to minimize the slice size.

Hint: Clean the blade between samples by stabbing into an unused gel area near the top.

- (4.) Transfer the gel slice to a clean microcentrifuge tube.

A > Fast flush extraction

- (1.) Place the gel slice into a spin column. Centrifuge at 15 000 × g for 1 min. Collect the flow-through.

Note: Most of the liquid will be extracted from the gel into the collection tube.

Hint: Alternatively, puncture the bottom of a 0.5 mL tube with a needle, add a small cotton plug, and place it inside a 1.5 mL tube. Spin at 5 000 × g for 5 min.

- (2.) Rotate the column 180° and spin again at 15 000 × g for 1 min.

Quality assurance: Use a blue light flashlight to check for residual fluorescence. If DNA remains, add 50 µL 5 mM Tris pH 8.0 and repeat the spin. Expect a faint greenish glow in the eluate. 

- (3.) Combine all flow-throughs. Discard the column.

- (4.) *Optional:* Further purify by silica spin column or ethanol precipitation  SOP0021. 

Note: The eluate can be used directly for ligation, PCR, sequencing, or probe labeling.

 [AMI+17]

B > Extraction with guanidinium thiocyanate

☐ Buffer QG (Extraction buffer), 0.5 mL <R>	☐ Isopropyl alcohol
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- (1.) Weigh the gel slice.
- (2.) Add 350 μ L Buffer QG per 100 mg gel. If the gel slice weighs less, supplement with electrophoresis buffer to 100 mg.
- (3.) Melt the agarose at 40–60 °C. Mix occasionally. Cool to room temperature.
- (4.) If the phenol red indicator turns red, add 10 μ L 3 M sodium acetate pH 5.0.

Critical: Do not skip this step—binding will fail at neutral or basic pH.

- (5.) Add 150 μ L isopropanol per 100 mg agarose.
- (6.) Bind DNA to silica  SOP0021. Skip Buffer PB.

Hint: If volume exceeds 700 μ L, reload after first spin. Silica membranes bind up to 25 μ g DNA.

Analyses

- Measure nucleic acid concentration and purity at 260 nm, 280 nm, and 230 nm.

Nucleic acid	A260 = 1.0	A260/A280	A260/A230
dsDNA	50 ng/ μ L	1.8–1.9	2.0–2.2

Note: Extinction coefficients and purity ratios depend on sequence. A260/A280 varies 0.2–0.3 with pH.

Quality assurance: High A230 absorbance indicates residual guanidine contamination.

Troubleshooting

Separation and excision of the desired DNA fragment from agarose gels

In Step 1:

- Fragment resolves into multiple bands after digestion.
 - Heat-inactivate the restriction enzyme or purify before gel loading. Some enzymes bind to DNA and alter its migration.

Extraction with guanidinium thiocyanate

In Step 5:

- Mixture appears cloudy or gelatinous
 - The gel may be incompletely dissolved. Proceed with column binding, then flush with 0.5 mL Buffer QG. Let stand 1 min before spinning.

Recipes

Buffer QG (Extraction buffer), pH 6.6

Amount	Ingredient	Stock	Final
100 g	Guanidine thiocyanate [593-84-0]	118.16 g/mol	5.5 M
3 mL	Tris-Cl, pH 7.4  R0055	1 M	20 mM
45 µL	Phenol red [143-74-8]	10 g/L	30 mg/L
To 154 mL	Water, reagent-grade		

Prepare Buffer QG in the bottle without weighing. Add 117 mL water per 100 g guanidinium thiocyanate, bring to 65 °C to dissolve. Adjust pH and final volume. Dispense into 15 mL aliquots.

Buffer QG (Extraction buffer)

5.5 M Guanidine thiocyanate, 20 mM Tris-Cl,
30 mg/L Phenol red, pH 6.6



DANGER

Serious eye damage

Date: Sign: R0135

List of references

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Change log

2021-04-01 Nick Coleman Original protocol.
2023-11-26 Benjamin C. Buchmuller Adaptation as SOP.

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